AIC04
Differential and Current Amplifiers

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Cover both Lab 4.A differential voltage amplifiers and Lab 4.B differential current amplifiers from *AIC Lab.pdf*, PDF pages 41–47. The model library is supplied separately by the user. The 1.0 V supply is a teaching adaptation, not a certification of the model voltage rating. Numerical plots for the handout's original 3.3 V process are not used directly as pass thresholds. The circuits are rebuilt from the handout topology; the earlier code is used only to review dimensions and omissions.

2 Specification Mapping

S denotes actual simulation and D derivation/discussion. D01 identifies the filename prefix; case numbering starts at C0.

Requirement and PDF pages	Deck / check	Status
Three differential voltage amplifiers (41–42)	D01–03	S+D
ICMR and device saturation checks (42)	D04; 21 points	S+D
Gain, GB, CMRR and PSRR (43–44)	D01–03; AC families	S+D
Equal-W/L scaling and cascode comparison (44)	D01/C4–6; D02–03	S+D
Three current amplifiers and 1 kOhm load (45–47)	D05	S+D
100/1000/10000 um widths and Rin/Rout (46–47)	D05/C1–12	S+D
PMOS source dimensions and tradeoffs (47)	D05/C13	S+D

3 Circuit and Design Derivation

$$A_d \simeq \frac{g_m}{g_{ds,n}+g_{ds,p}}, \qquad f_u \simeq \frac{g_m}{2\pi C_L}, \qquad \mathrm{CMRR}_{dB} = 20\log_{10}\left|\frac{A_d}{A_{cm}}\right|.$$

$$\mathrm{PSRR}_{\pm,dB} = 20\log_{10}\left|\frac{A_d}{A_{supply,\pm}}\right|, \quad R_{in} \simeq 1/g_{m,in}, \quad A_i \simeq \frac{(W/L)_{out}}{(W/L)_{in}}.$$

Voltage amplifiers use $C_L = 10$ pF. The basic reference is 50 μA; cascode baseline comparisons explicitly use 0.5 μA to retain 1 V headroom, with 50 μA cases also retained. M1/M2 use 425/0.3 μm, PMOS loads 1350/0.3 μm, and reference/tail devices 85/0.3 and 870/0.3 μm. D01/C4,C5 scale both

W and L of input/load devices by 3,5 without changing the tail mirror. Full-cascode extra NMOS gates are held at 0.6 V.

The current amplifier deck contains simple (s), cascode (c) and differential (d) versions. Reference NMOS width is 50 μm; output width is 100/1000/10000 μm; $L = 0.3\,\mu$m. A real 1 kΩ load is grounded. C0 retains matched source/sink currents, potentially driving the output near ground and outside its operating region. C1–3 explicitly add nominal 0.5 mA PMOS bias to place the load near 0.5 V. C7–9 change both lower cascode widths. C13 doubles s/c PMOS source widths; the first d PMOS also changes but its other two remain ratio-defined, making d an additional imbalance counterexample, not a matched comparison.

4 Simulation Setup

Runs use PrimeSim HSPICE S-2021.09 win64, a baseline temperature of 27 °C, `runlvl=5`, `accurate`, and `.op/.print/.measure` outputs. Every `.alter` resets all baseline parameters and temperature to prevent state from carrying into the next case. Actual runs completed without an error or failed measurement. Suppressed model warnings remain, so a warning-free run is not claimed. Public netlists contain no model paths. This comment marks the model setup required locally before execution; the model and its configuration remain local:

```
* MODEL_SETUP_REQUIRED
```

CSV files and numerical summaries are in the lab's `results/` directory. Every figure is redrawn from actual printed data, not a waveform-viewer screenshot. `manifest.json` supplies specification mapping and case definitions.Five decks contain 52 cases. AC sweeps 1 Hz–10 GHz with 60 points/decade. Differential voltage is $+0.5/-0.5$ V, hence 1 V total; common mode drives both inputs with 1 V. Supply injection holds ideal inputs relative to global ground. Differential current uses $+0.5/-0.5$ A. ICMR sweeps 0–1 V in 50 mV steps with device V_{DS} and V_{DSAT} stored at every point. Rout cases disable input AC and inject 1 A, using a huge series load inductor to retain DC bias while removing the 1 kΩ AC load. This is a measurement fixture, not a realizable inductor.

5 Results and Specification Checks

Case	A_d (dB)	f_{3dB} (kHz)	f_u (MHz)	V_o (V)
D01/C0	39.60	620.351	59.217	0.7002
D01/C4	45.44	197.286	32.930	0.6921
D01/C5	46.53	125.985	22.035	0.6907
D01/C6	32.58	21.771	0.874	0.8885
D02/C0	46.82	4.259	0.750	0.7530
D02/C4	41.17	608.640	55.649	0.3427
D03/C0	68.26	0.349	0.730	0.7523
D03/C4	54.58	125.026	54.085	0.3428

Topology	CMRR (dB)	PSRR+ (dB)	PSRR- (dB)
Basic	81.18	39.60	81.14
Load cascode	87.03	46.82	86.73
Full cascode	130.18	68.26	108.07

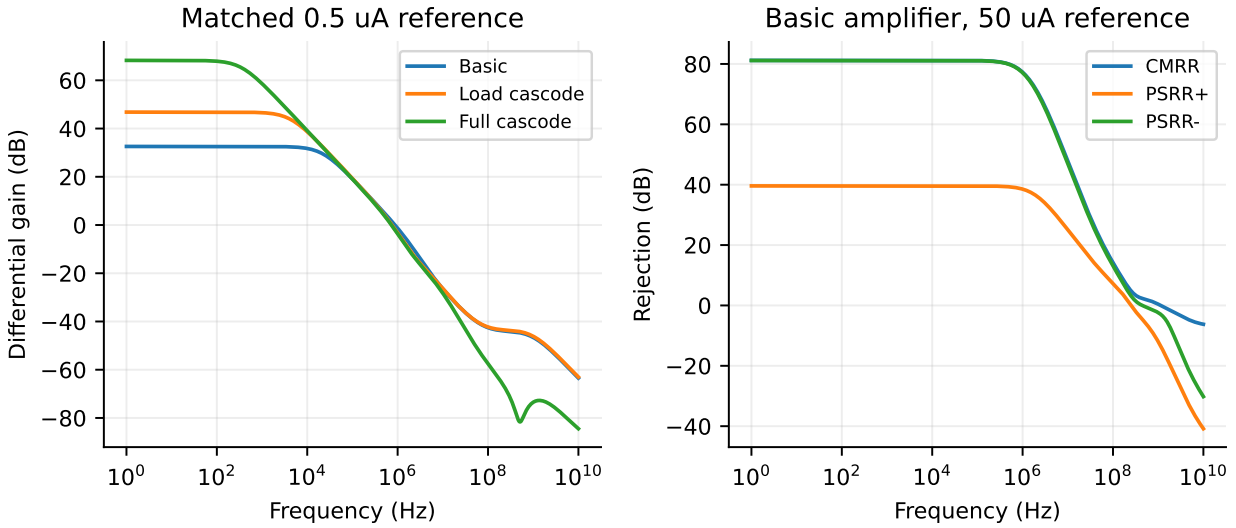


Figure 1: Left: three topologies at matched 0.5 microamp reference. Right: rejection for the 50 microamp basic amplifier; the panel biases differ.

For the common 0.5 μA scan, contiguous sampled intervals with every device satisfying $|V_{DS}| \geq |V_{DSAT}|$ are 0.20–1.00 V (basic), 0.20–0.95 V (load cascode) and 0.20–0.55 V (full cascode). Endpoint resolution is 50 mV, and the basic upper bound reaches the sweep limit. This criterion does not establish strong inversion or constant transconductance.

Case/Type	$\lvert A_i \rvert$	R_{in} (kΩ)	R_o (kΩ)	f_{3dB} (MHz)	V_o (V)
C0/s	0.501	4.061	–	112.679	0.013
C0/c	0.503	9.127	–	340.870	0.014
C0/d	0.731	7.958	–	49.273	0.013
C1/s	1.916	4.061	15.814	0.778	0.512
C1/c	1.864	9.133	16.392	0.792	0.514
C1/d	2.797	7.958	15.814	0.794	0.512
C2/s	18.415	4.059	9.522	0.459	0.502
C2/c	18.271	9.134	12.146	0.443	0.515
C2/d	158.918	36.990	9.525	0.440	0.502
C3/s	127.907	4.009	1.744	0.025	0.436
C3/c	150.941	9.135	3.316	0.022	0.534
C3/d	1511.730	121.390	0.271	0.022	0.082
C13/s	0.751	2.114	–	0.302	0.921
C13/c	0.678	4.770	–	0.319	0.923
C13/d	0.007	4.192	–	290.717	0.533

Lower width (μm)	$\lvert A_i \rvert$	R_{in} (kΩ)	R_o (kΩ)	f_{3dB} (MHz)
100	0.935	8.949	16.337	0.810
1000	0.936	8.528	16.337	0.793
10000	0.895	7.674	16.325	0.533

The s/c/d labels denote simple/cascode/differential current amplifiers. Failed operating regions in C0, imbalance in C13 and actual output voltages remain in the results. Values disagreeing with square-law predictions are retained. A dash means Rout was not measured independently for that case. Full CSV data preserve the frequency dependence.

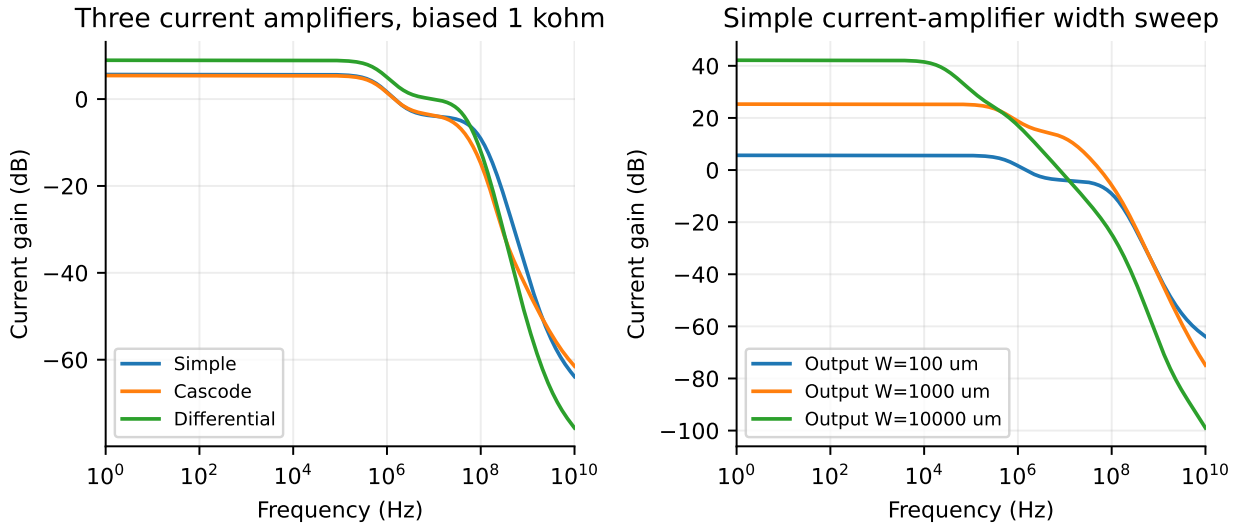


Figure 2: Left: the three current amplifiers in C1. Right: simple-amplifier output-width sweep including actual bias and loading.

6 Discussion

Compared with a single-ended CS, differential inputs reject common disturbances and support negative feedback, but require more devices, bias current, matching and common-mode headroom. Increasing device length at equal W/L reduces output conductance: basic gain rises from 39.60 dB to 45.44/46.53 dB while bandwidth decreases. Cascoding raises output impedance and gain at the cost of headroom; full cascode has the narrowest ICMR at matched low current. Rejection assumes nominal matching, without mismatch or silicon guarantees.

The simple current mirror needs less headroom and capacitance. Cascode internal nodes and body effect can raise input impedance while reducing loaded current gain. Larger output W and bias increase ideal mirror ratio but also capacitance and output conductance, so Rin, Rout and GB do not improve together. PMOS sizing changes DC balance: C13 approaches the upper rail and must be interpreted with its bias. The differential current topology accepts two inputs but accumulates errors and capacitance across mirror stages. The text, width tables and plot names on pages 46–47 are not fully consistent; the explicitly listed devices and cases define this report, without imposing the old numbers.

7 Conclusion

Every topology, measurement and discussion in Lab 4.A/B is represented. Actual results show gain, rejection and operating-region tradeoffs, including failed bias cases. Original-process numbers and manufacturing validation are not claimed.

8 MOS transistor schematic

These editable vector schematics connect equal node names within a circuit. Primitive-cell names are local pins; the hierarchy defines their connections. B denotes the body terminal; 0 is global ground. Each panel identifies its netlist and scope; measurement sources are connected as described. Complete six-MOS baseline differential amplifier from `01_basic_differential.sp`.

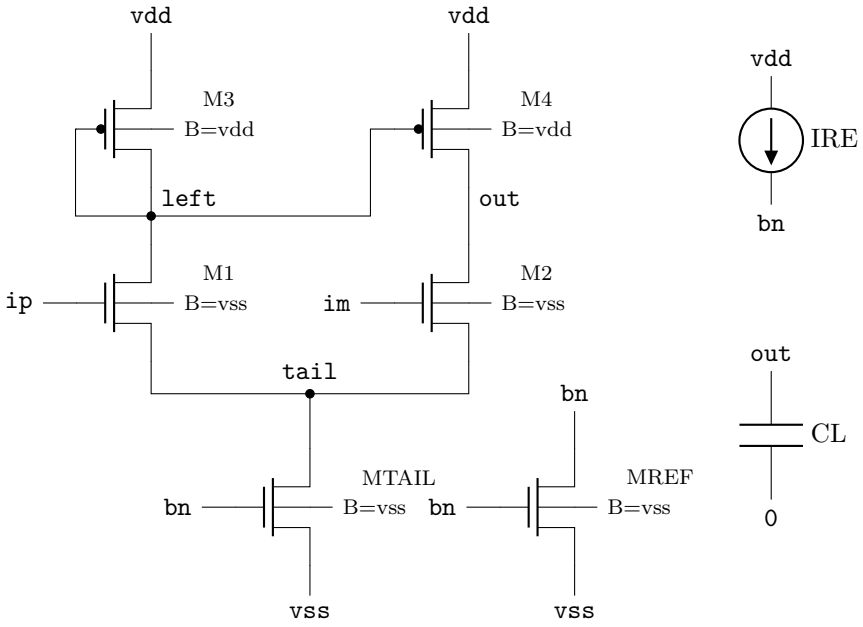


Figure 3: NMOS differential pair, PMOS mirror load and NMOS tail-current mirror.

All NMOS bodies connect to vss, including M1/M2 whose sources are tail. All PMOS sources/bodies connect to vdd. At scale=1 the widths are MREF=85, MTAIL=870, M1=M2=425 and M3=M4=1350 μm; all lengths are 0.3 μm. The scale parameter multiplies both W and L of M1–M4 only. IRE is 50 μA, CL is 10 pF. The input sources impose $ip = cmnode + diff/2$ and $im = cmnode - diff/2$. vdd=1 V and vss=0 V at DC; CL returns to global 0, which is distinct from an AC-perturbed vss in rejection tests. Cascode-load, full-cascode and current-amplifier decks are separate variants; the figure covers the named baseline deck.